

Hugh Palin

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EDUCATION

University of Illinois at Urbana-Champaign

B.S in Computer Engineering, Minor in Business

Aug. 2022 – May 2026

GPA: 3.62/4.00

EXPERIENCE

Lead Instructor

iD Tech at Northwestern University

May 2025 – Aug. 2025

Evanston, IL

- Taught high school students AI in Python using TensorFlow to train image recognition models on real-world inputs
- Instructed 96 students in C++ for VEX robotics, covering motor control, sensors, and mechanical systems
- Led a team of 5-6 instructors, mentored new instructors, and managed daily classroom operations at iD Tech

Undergraduate Research Assistant

Northwestern University

May 2024 – Aug. 2024

Evanston, IL

- Conducted research focused on identifying pathways to catalyze reactions yielding value-added products, such as pharmaceuticals or biofuels. Leveraged computational modeling to evaluate the feasibility of reactions using HPC
- Designed a Python workflow to automate the generation of molecular structures for simulating enzymatic reactions, reducing manual input by 99%. Ensured seamless integration with downstream machine learning models

Database Engineer Intern

SC Johnson & Son, Inc.

May 2023 – Aug. 2023

Racine, WI

- Developed a Python-based digital cataloging system converting paper reports into a structured SQL database, automating data extraction, enhancing accessibility, visualization, and improved query performance by 40%
- Optimized full-text SQL search with metadata indexing, enabling fast, keyword-based document retrieval

PROJECTS

Suction Sense | C++, Python

Aug. 2025 – Present

- Developed system to detect excess OR suction use—cutting waste, wear, and saving hospitals thousands yearly
- Developed ESP32 embedded firmware to process sensor data and transmit telemetry via MQTT to a Raspberry Pi
- Built a C++ web application using Crow and SQLite to visualize real-time data with a responsive touchscreen
- Designed a PCB integrating sensors, battery management system, and dual power inputs for continuous operation
- Implemented data collection and verification ensuring $\geq 99\%$ reliability in telemetry and $\leq 2s$ UI refresh rate

Android Weather App | Java, Android Studio, SQLite, Gemini API

Aug. 2025 – Present

- Built a full-stack Android app with user authentication, dynamic UI customization, and persistent state storage
- Implemented login with Android AccountManager for secure authentication and personalized user sessions
- Built a Content Provider-based SQLite database for CRUD operations to manage user-specific city locations
- Integrated LLM-driven UI themes using the Gemini API to dynamically generate color schemes from text input.
- Collaborated within an Agile team using GitHub for commits, issue tracking, and weekly progress documentation
- Deployed and tested on an Android emulator (API 29, Pixel 3a) to validate reliability and multi-user functionality

Custom RISC-V Operating System | RISC-V, C

Mar. 2025 – May. 2025

- Designed a custom operating system with a shell interface to interact with a file system and launch user programs
- Built core OS components including scheduling, memory management, and device drivers with UART and virtio, incorporating support for interrupts and memory-mapped I/O in a large codebase
- Implemented process scheduling and forking and context switching to support multi-tasking across user programs
- Integrated a filesystem with multi-process and file descriptor support across isolated memory spaces
- Developed memory management modules with paging and virtual memory to allocate resources efficiently

Convolutional Neural Network | C

Oct. 2024 – Nov. 2024

- Implemented forward propagation of a LeNet-5 CNN for accurate handwritten digit recognition using ML methods
- Boosted training and testing speed with CUDA parallelism on an A40 GPU using cuBLAS and shared memory

TECHNICAL SKILLS & COURSEWORK

Languages: C, C++, Python, CUDA, SystemVerilog, VHDL, RISC-V assembly, JavaScript, Java, HTML/CSS, SQL

Courses: Data Structures & Algorithms, Discrete Structures, Differential Equations, Machine Learning, Computer Systems Engineering, Embedded Systems Design, Operating Systems Design, Software Engineering I